

E Albert Einstein Cup 2024

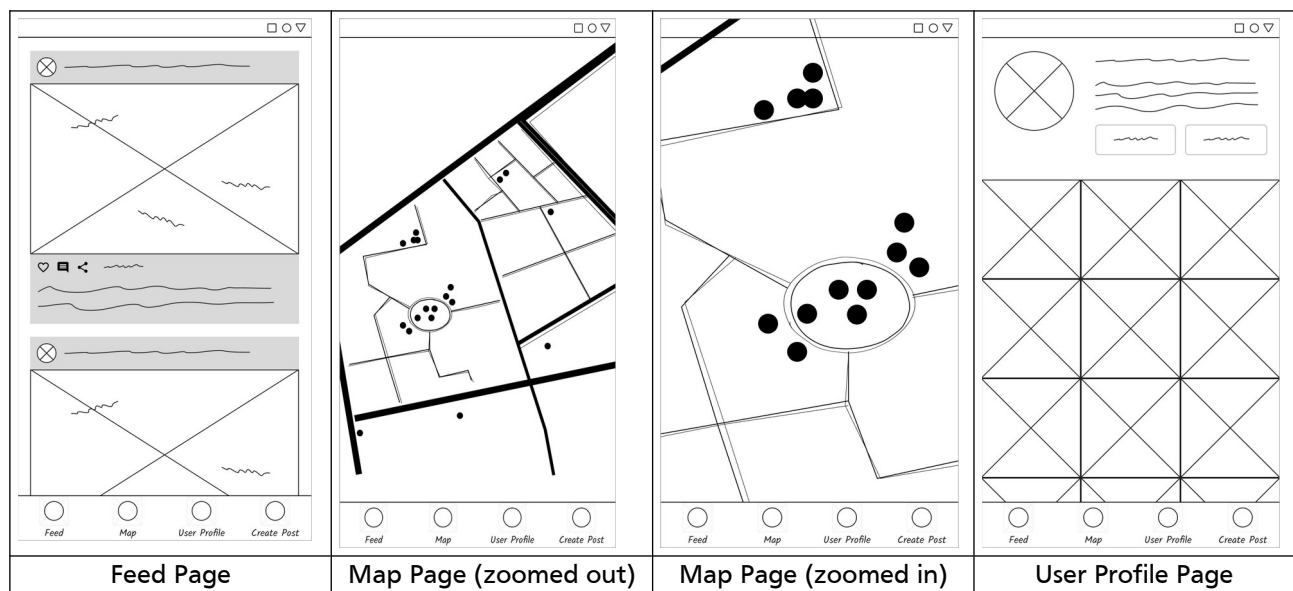
Session 3: Functionality

Introduction

The product manager has provided you with a description and some wireframes for the phone app. As a mobile app developer, your task is to develop a working app, offering the functionality described in this document.

Description of project and tasks

You do not need to consider the aesthetics of the program interface. The work can be inconsistent with the given wireframes (which are incomplete), but you need to properly consider the ease of use. You have **3.5 hours** to implement the app for the provided phone.



General Demands

1. All text with quotation marks must be the same in your app.
2. The app may only be used in portrait mode.
3. The app consists of three main pages:
 1. Feed Page: Displays a collection of posts shared by other users.
 2. Map Page: Presents user posts as markers on Google Maps.

E Albert Einstein Cup 2024

3. User Profile Page: Displays the profile of a user. If the profile belongs to the user who is currently logged in, they have the ability to make edits to their profile.
4. The three main pages are reachable from a bottom navigation bar that's shown on all pages except the authentication page.
5. Connect to the API at <http://api.skills.local:1308x>, where x is your Workstation number.
6. To access the main pages, authentication is required. The authentication page should consist of a simple username input (no password). The username must be a non-empty string.
 1. The authentication state (whether the user is signed in and with which username) should be persisted across restarts of the app.
7. If any user has no profile picture, use the provided file *Default Profile Picture.png* as a fallback.

Page Demands

1 Feed Page

Elements

The Feed Page displays posts from all users except yourself. Each post includes the photo, stickers, like count, caption, and the author's name and profile picture. Additionally, it should provide user interface elements to like, comment on, and share the post.

Functional Requirements

1. Display posts from all users except yourself in reverse chronological order (most recent is at the top).
2. Post images are decorated with their stickers (small images on top of the post's main image) using correct coordinates and rotation (see "New Post Demands").
3. Clicking on the author's name or profile picture navigates to their profile page.
4. The like button on a post toggles whether the current user likes that post: Upon tapping the like button, the like count increases. If the user taps the like button again, the like gets removed. Both actions should update the API and also update the like count shown to the user. Additionally, the like button should visually indicate whether the user likes the post or not.
5. The comment button opens a bottom sheet showcasing the post's comments and provides an option to add a new comment.
6. The share button enables users to share the post with external applications using the system's share dialog, e.g., to the gallery or by email. The shared content should include the post's photo and stickers as well as its caption and author's name.

E Albert Einstein Cup 2024

2 Map Page

Elements

The Map Page displays all posts using markers on an embedded map using Google Maps. The size of these markers adjusts according to the map's zoom level: Markers appear as small dots when zoomed out and become larger dots as you zoom in.

Functional Requirements

1. Display all posts as markers on a map.
2. Adjust the size of the markers based on the map's zoom level:
 - When zoomed out, markers should appear as small dots.
 - When zoomed in, markers should transform into larger dots.
3. When a user clicks on a marker, the corresponding post should be displayed in the Post Page.

3 User Profile Page

Elements

Display the user's profile picture, name, and profile caption at the top of the page. Below, present a grid featuring all the user's posts, each only with its photo and stickers. If the profile is that of the currently logged-in user, include an "Edit" and a "Sign out" button.

Functional Requirements

1. Display the user's profile picture, name, and profile caption prominently.
2. Present a scrollable three-column grid showcasing all of the user's posts (only the photos and stickers need to be shown) in reverse chronological order (most recent is at the top). Each post preview in this grid should be square, and some parts of the post's photo and stickers may therefore be hidden.
3. Make each post in the grid clickable, opening the post in the Post Page for detailed viewing upon selection.
4. If the profile being viewed belongs to the currently logged-in user:
 1. Allow editing the profile caption via the "Edit" button.
 2. Allow changing the profile picture to a new one from camera or gallery by tapping it.
 3. Allow the user to sign out via the "Sign out" button.

E Albert Einstein Cup 2024

4 Post Page

Elements

Display one post, including its photo, stickers, like count, author's name and profile picture, along with a caption. Also show all comments directly on this page. It should also provide user interface elements to like, comment on, and share the post.

Functional Requirements

1. The post image is decorated with stickers using correct coordinates and rotation (see "New Post Demands").
2. Clicking on the author's name or profile picture will navigate to their profile page.
3. The like and share buttons work the same as on the Feed Page.
4. The comment button provides an option to add a new comment.

New Post Demands

Elements

Each main page must include a button for creating a new post.

Functional Requirements

1. Include a "Create New Post" button on every main page.
2. Offer users the option to either take a new picture using the camera or select one from their gallery.
 - When a new picture is taken with the camera, automatically use the device's current location as the geolocation for the Map Page.
 - When selecting a picture from the gallery, the user must specify the location by placing a marker on a map.
3. Provide functionality for users to add a caption to their post.
4. Enable users to decorate their posts with stickers:
 1. The user has an overview of all stickers provided by the API and can drag any of them onto the photo. Each sticker can be used multiple times.
 1. The sticker's side length should be 20 % of the shorter side of the post photo.
 2. The user should be able to drag an added sticker to somewhere else on the post's photo.
 1. In the API, the sticker's position is stored as X, Y coordinates (for horizontal and vertical, respectively), each ranging from 0 (top left) to 1 (bottom right). These coordinates mark the centre of the sticker.

E Albert Einstein Cup 2024

3. The user should have the option to rotate the stickers to their desired orientation.
 1. In the API, this is stored as a rotation around the sticker's centre in radians ranging from 0 (inclusive) to two pi (exclusive). For example, 0 is the normal direction, $\frac{\pi}{2}$ is rotated 90° clockwise, π is upside down / rotated 180°, and $\frac{3\pi}{2}$ is rotated 90° counter-clockwise. However, the rotation is not limited to these four values.
4. The user should have the possibility to delete a sticker, e.g., if they added it accidentally.
5. After a post is created, it should automatically appear in the proper pages. E.g., it should appear on the User Profile Page without the user having to refresh it.

Submission

Your submission should contain the following top-level files and directories:

- the compiled APK
- a directory with the full source code
- optional: a README file containing notes for the marking

These files and directories should then be pushed to the remote Git repository.